Lecture 14

Degree, Poincaré–Hopf and de Rham's theorem

Objectives. *The integral of a normalized top form assigns a number to a smooth map between nonempty closed connected oriented manifolds of the same positive dimension. We prove that this number, the degree, is an integer and is unchanged under homotopy. Degree obstructs a nonvanishing smooth tangent field on an even sphere and proves the smooth fixed-point theorem. Two cited theorems then relate vector-field zeros and periods to the homology of Part I.*

By the end of this lecture you will be able to:

- Define degree with its dimension, orientation and compactness hypotheses.
- Compute degree as a signed count over a regular value.
- Prove the even-sphere hairy-ball theorem for smooth fields and the smooth Brouwer theorem.
- Compute the two local indices of the specified field on S^2.

14.1 Integration and top cohomology

Recall that a closed manifold is compact and has no boundary. All manifolds below are smooth. Cohomology has real coefficients. On an oriented n-manifold every n-form is closed, since all $(n+1)$-forms vanish. For a closed manifold, Stokes makes integration zero on exact forms. Integration therefore defines a linear map on $H^n(M)$.

Theorem 14.1 (Top cohomology · *sketched*)**.** *For a nonempty closed connected oriented n-manifold M,*

$$I_M \colon H^n(M) \longrightarrow \mathbb{R}, \qquad I_M[\alpha] = \int_M \alpha$$

is an isomorphism. Thus two top forms have equal integrals exactly when they differ by an exact form.

Sketch. For $n = 0$, connected M is a point and integration is its signed value, an isomorphism. For $n \geq 1$, a positive bump function in one positive chart gives a top form of positive integral. Rescaling it gives integral one, proving surjectivity. For injectivity the local input is this compact-support version of the Poincaré lemma: on $\mathbb{R}^n$, a compactly supported top form of integral zero has a compactly supported primitive.

The local construction uses induction and the fundamental theorem of calculus. For $n = 1$, integrate its coefficient from $-\infty$; zero total integral makes the primitive zero also beyond the support. For $n > 1$, write the coefficient as $f(x', u)$, set $h(x') = \int_{\mathbb{R}} f(x', u)\, du$, and

choose a compactly supported smooth function $\rho(u)$ of integral one. Split the coefficient as $(f - h\rho) + h\rho$. The first term has zero integral in every u fibre, so integrating in u gives a compactly supported primitive, with the sign from moving $\mathrm{d}u$ past $\mathrm{d}x'$. The second term has a primitive by the induction hypothesis applied to $h\mathrm{d}x'$, whose integral is $\int f = 0$, and then wedged with $\rho\mathrm{d}u$.

On M, choose a finite cover by positive coordinate balls diffeomorphic to $\mathbb{R}^n$. A partition of unity splits a top form into ball-supported pieces. Connectedness permits ordering the balls so that each after the first meets an earlier ball. In each chosen overlap take a top form of integral one. Starting with the last piece, subtract its integral times this overlap form and add the same form to the earlier piece. Repeating backwards makes every piece have integral zero, since the total is zero. The local construction now supplies primitives which extend by zero and sum to a global primitive. The cover and transfer details are the sketched steps. See Lee, *Introduction to Smooth Manifolds*, second edition, Theorems 17.30–17.31, pp. 454–455. ∎

For example, on outward S^2, the form $\beta/(4\pi)$ from Lecture 13 has integral one. A 2-form α of integral $a \in \mathbb{R}$ consequently satisfies $[\alpha] = a[\beta/(4\pi)]$.

14.2 The degree of a map

Let M and N be nonempty closed connected oriented manifolds of the same dimension $n \geq 1$. Let $f\colon M \to N$ be smooth. Choose a top form ν on N with integral one. The *degree* of f is

$$\deg f = \int_M f^*\nu.$$

It compares the orientations and will count inverse images with signs.

Theorem 14.2 (Degree and homotopy · *proved given top cohomology and cited smoothing*)**.** *The degree is independent of ν. For every top form α on N,*

$$\int_M f^*\alpha = (\deg f) \int_N \alpha.$$

Homotopic smooth maps have the same degree. The identity has degree one, and a constant map has degree zero when $n \geq 1$.

Proof. Two choices of ν differ by $\mathrm{d}\eta$, by the top-cohomology theorem. Their pullbacks differ by $\mathrm{d}f^*\eta$, whose integral is zero by Stokes. For an arbitrary α, write $a = \int_N \alpha$. The form $\alpha - a\nu$ has integral zero, so it is exact. Pulling back and integrating gives the claimed formula.

For smoothly homotopic maps, the homotopy operator of Proposition 13.1 makes the pullbacks of ν differ by an exact form. Their integrals coincide. A continuous homotopy between smooth maps can be smoothed by Lee's Theorem 6.29, recalled in Lecture 13, so the same conclusion holds. The identity pulls ν back to itself. A constant map has zero differential, hence pulls a positive-degree form back to zero. ∎

For example, the reflection $(x, y) \mapsto (x, -y)$ on S^1 reverses the angular form, so its degree is -1. The degree also multiplies under composition. For a smooth map g from N to another such manifold of the same dimension, choose a top form ν of integral one on its target. Apply the integral formula to $g^*\nu$ and then to ν to obtain $\deg(g \circ f) = (\deg g)(\deg f)$.

Closedness matters. The map $x \mapsto x/2$ from $[0,1]$ to itself pulls $\mathrm{d}x$ back to $\mathrm{d}x/2$. The quotient of the two integrals is $1/2$, but the interval has boundary, so this quotient is not a degree under our definition.

14.3 Integrality and regular values

A point $y \in N$ is a *regular value* of f if $\mathrm{d}f_x \colon T_xM \to T_yN$ is surjective for every $x \in f^{-1}(y)$. An empty inverse image satisfies this condition. In equal dimensions each such differential is invertible. Its *local degree* is $+1$ if it preserves orientation and -1 if it reverses orientation. Write this sign as $\operatorname{sgn}\det \mathrm{d}f_x$, using positive bases in both tangent spaces. An inverse branch is a neighbourhood mapped diffeomorphically onto a small target neighbourhood W. The diagram names three such branches V_1, V_2 and V_3 and shows their signs. The proof below constructs the common neighbourhood W.

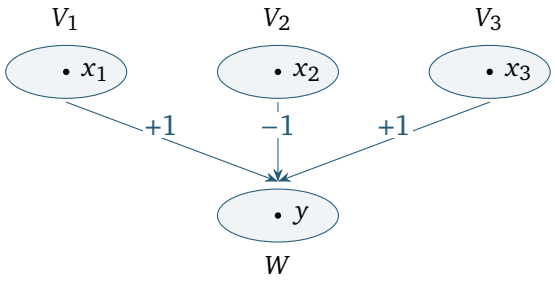


Figure 14.1: Three inverse branches over a small neighbourhood W contribute signs $+1$, -1 and $+1$. Their total contribution is one.

Proposition 14.3 (Signed count · *proved given top cohomology; regular values exist by cited Sard*)**.** *For every regular value $y \in N$, the inverse image is finite and*

$$\deg f = \sum_{x \in f^{-1}(y)} \operatorname{sgn}\det \mathrm{d}f_x .$$

In particular the degree is an integer, independent of the chosen regular value.

Proof. The inverse function theorem isolates each inverse-image point. The fibre is closed in compact M, hence compact. Its isolating neighbourhoods give an open cover with a finite subcover, so the fibre is finite. Write its points as $x_1, \dots, x_m$ when it is nonempty. Choose disjoint neighbourhoods U_i on which f is a diffeomorphism onto a neighbourhood of y. The complement $K = M \setminus \bigcup_i U_i$ is compact and has no point mapping to y. Its image is compact and closed. Choose a small connected coordinate ball W about y inside every $f(U_i)$ and disjoint from $f(K)$. The sets $V_i = U_i \cap f^{-1}(W)$ are then all the inverse branches, each diffeomorphic to W. On a connected branch the orientation sign is constant, since its nonzero Jacobian determinant is continuous.

Choose a smooth top form ν supported in W with integral one. Change of variables on each branch gives

$$\int_M f^*\nu = \sum_{i=1}^{m} \int_{V_i} f^*\nu = \sum_{i=1}^{m} \operatorname{sgn}\det \mathrm{d}f_{x_i} .$$

If the fibre is empty, choose W disjoint from the compact image $f(M)$. The pullback of such a supported ν is zero, giving degree zero and the empty sum. This proves the formula for every regular value.

Sard's theorem says that the critical values of a smooth map have measure zero in the target. We cite Lee, Theorem 6.10, p. 129. Here a critical value is the image of a point with nonsurjective differential. A nonempty coordinate ball has positive measure, so regular values exist. The formula at any of them proves integrality. Since it computes the already defined degree, it also proves independence of the regular value. ∎

14.4 Antipodal maps, tangent fields and fixed points

A fixed point of a map f means a point x with $f(x) = x$. Let μ be the outward area form on S^n, characterized in Lecture 12 by $\mu_p(v_1, \dots, v_n) = \det[p\ v_1\ \cdots\ v_n]$.

Corollary 14.4 (Applications of degree · *proved given top cohomology*)**.** *For $n \geq 1$, the antipodal map $a\colon S^n \to S^n$, given by $a(x) = -x$, has degree $(-1)^{n+1}$. Every smooth tangent field on S^{2m}, with $m \geq 1$, has a zero. Every smooth map from the closed unit ball $\overline{B}^d$ to itself has a fixed point, for $d \geq 1$.*

A nonvanishing unit tangent field would give the semicircle in the figure at every sphere point.

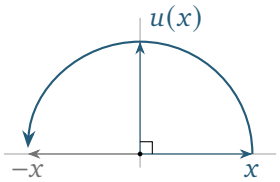


Figure 14.2: In the plane spanned by the orthogonal unit vectors x and $u(x)$, the unit semicircle joins x to $-x$ through $u(x)$.

Proof. The derivative of the antipodal map sends every tangent vector to its negative. Thus $a^*\mu = (-1)^{n+1}\mu$, because the determinant has $n+1$ columns. The integral definition gives its degree.

Suppose a tangent field X on S^{2m} never vanishes and normalize it to $u(x) = X_x/\|X_x\|$. Tangency gives $x \cdot u(x) = 0$. The map $H\colon [0,1] \times S^{2m} \to S^{2m}$ given by

$$H(t,x) = \cos(\pi t)x + \sin(\pi t)u(x)$$

has squared norm $\cos^2(\pi t) + \sin^2(\pi t) = 1$. It is a smooth homotopy from the identity to the antipodal map. Their degrees would be equal, contradicting $1 \neq -1$. This proves the *hairy-ball theorem* for even spheres.

For the ball of dimension $d \geq 2$, suppose a smooth f has no fixed point. The map $G\colon \overline{B}^d \to S^{d-1}$ given by $G(x) = (x - f(x))/\|x - f(x)\|$ is smooth. Its boundary restriction g has degree zero: for a normalized top form ν on S^{d-1}, Stokes gives $\int_{S^{d-1}} g^*\nu = \int_{\overline{B}^d} \mathrm{d}G^*\nu = 0$, since $\mathrm{d}\nu = 0$. On the boundary the formula

$$H(t,x) = \frac{x - tf(x)}{\|x - tf(x)\|}$$

is a smooth homotopy from the identity to g. For $t < 1$ its denominator is nonzero because $\|tf(x)\| \leq t < 1 = \|x\|$. At $t = 1$ it is nonzero by the assumption. Homotopy invariance gives $\deg g = 1$, a contradiction. For $d = 1$, the function $f(x) - x$ is nonnegative at -1 and nonpositive at 1, so the intermediate value theorem gives a zero. ∎

The fixed-point conclusion is the *Brouwer theorem*. It also holds for continuous maps. That extension is cited here from Lee, Theorem 17.39, pp. 459–460, for dimensions $d \geq 2$. Its proof uses Theorem 17.38 and smooth approximation into manifolds, Theorem 6.26. For $d = 1$ the intermediate value argument above applies directly to continuous maps. Sheet 14 develops the approximation argument.

14.5 The index of a vector-field zero

An isolated zero of a vector field is a zero with a neighbourhood containing no other zero. In dimension $n \geq 2$, choose coordinates centred at such a zero p and a small outward-oriented coordinate sphere S_ε^{n-1}. Write the coordinate components of the field as $X(x) \in \mathbb{R}^n$. Its *index* is

$$\operatorname{ind}_p X = \deg\left(x \longmapsto \frac{X(x)}{\|X(x)\|} : S_\varepsilon^{n-1} \longrightarrow S^{n-1}\right).$$

Changing the small radius gives homotopic maps through spheres avoiding zeros. Independence of the coordinate chart is cited from Milnor, *Topology from the Differentiable Viewpoint*, §6, Lemma 1, pp. 33–35. In dimension one use half the difference of the component's signs at the right and left endpoints of a small coordinate interval centred at the zero. In dimension zero each point has index one. For a closed manifold define its Betti numbers by $b_k(M) = \dim_{\mathbb{R}} H_k(M;\mathbb{R})$ and its Euler characteristic by $\chi(M) = \sum_{k=0}^{\dim M} (-1)^k b_k(M)$. These dimensions are finite by the cited finite-dimensionality of de Rham groups (Lee, Problem 18-8, p. 489, an exercise with hints) and the comparison below. For a finite Δ-complex model, Theorems 6.2 and 6.3 identify this with the count of Lecture 6.

Theorem 14.5 (Poincaré–Hopf · *cited*). *For a smooth vector field with isolated zeros on a closed manifold M,*

$$\sum_{p : X_p = 0} \operatorname{ind}_p X = \chi(M).$$

Here χ is the Euler characteristic. The theorem is cited from Milnor, §6, p. 35.

On the unit sphere write $p = (x, y, z)$ and take $X_p = e_3 - z p$. This field provides the explicit test in the figure.

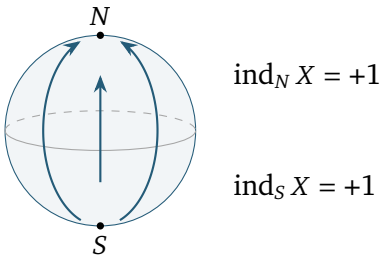


Figure 14.3: The field $X_p = e_3 - z p$ on S^2 points toward the north pole. It vanishes at the north pole N and the south pole S. Each zero has index +1.

The field is tangent because $p \cdot X_p = z - z\,\|p\|^2 = 0$. A zero requires $e_3 = z p$, hence $p = \pm e_3$. Near either pole use the coordinates $p = (x, y, \sigma\sqrt{1 - x^2 - y^2})$, where $\sigma = +1$ at the north pole and $\sigma = -1$ at the south pole. The coordinate components are $(-zx, -zy)$. On the small circle $(x, y) = \varepsilon(\cos\phi, \sin\phi)$, their normalization is $-\sigma(\cos\phi, \sin\phi)$. Both signs give one counter-clockwise turn as ϕ runs from 0 to 2π. Thus both indices are +1, and their

sum is $2 = \chi(S^2)$ from Lecture 6. A sink in two dimensions has index +1 here, not −1. For comparison, the planar source (x, y) has index +1, while the saddle $(x, -y)$ normalizes to $(\cos\phi, -\sin\phi)$ and has index −1.

14.6 Periods and singular homology

Recall from Lecture 6 that a singular cycle is a chain with zero boundary, and that $H_k(M;\mathbb{R})$ denotes homology with real coefficients. For a smooth singular simplex $\sigma\colon \Delta^k \to M$, integration means $\int_{\Delta^k} \sigma^*\alpha$, extended linearly to chains. A smooth simplex here extends smoothly to a neighbourhood of its parameter simplex. Arbitrary continuous singular simplices cannot be integrated this way.

Theorem 14.6 (De Rham · *cited*)**.** *For every smooth manifold M and $k \geq 0$, integration induces an isomorphism*

$$H^k(M) \longrightarrow \mathrm{Hom}_{\mathbb{Z}}(H_k(M;\mathbb{Z}),\mathbb{R}), \qquad [\alpha] \longmapsto \left([c] \longmapsto \int_c \alpha\right).$$

Here $\mathrm{Hom}_{\mathbb{Z}}$ *means homomorphisms of abelian groups. This includes replacing continuous cycles by smooth representatives. The cited construction uses Stokes on simplices, including their corners, and shows independence of all representatives. See Lee, Theorem 18.14, p. 484, with the comparison construction in Chapter 18.*

Thus the periods over all cycles determine a cohomology class completely. The same target is the real dual of $H_k(M;\mathbb{R})$. Indeed, real singular cochains are real-linear functionals on the real chain complex, equivalently additive real-valued functionals on integer chains.

A closed cochain vanishes on boundaries, so it restricts to a functional on homology. Every such functional extends from cycles to all chains by choosing a basis. A closed cochain restricts to zero on cycles exactly when it factors through the boundary map. Extending that factor to the preceding chain space makes it a coboundary.

For integer chains the same restriction and factorization argument uses a cited extension property in place of basis extension: every homomorphism from a subgroup of an abelian group to $\mathbb{R}$ extends to the whole group. This property is called injectivity of $\mathbb{R}$ and is stated in Lee, p. 473. Thus both homology duals compute the cohomology of the same real-valued cochain complex. This proves their identification given that cited property.

In finite dimension it gives $\dim H^k(M) = b_k(M)$. The real coefficients matter: an integer homology class annihilated by $m \geq 1$ has period P with $mP = 0$, hence $P = 0$. Differential forms therefore do not recover torsion classes.

14.7 Radial flux and complex power maps

Let $K \subset \mathbb{R}^3$ be a compact smooth region with connected boundary $S = \partial K$, oriented outward, and assume $0 \notin S$. The *radial Gauss map* $G\colon S \to S^2$ is given by $G(x) = x/\|x\|$. It records the direction from the origin. This map differs from the map assigning the outward unit normal to a surface point. The figure shows the two positions of the origin.



$0 \in \operatorname{int} K$ $\qquad$ $0 \notin K$

Figure 14.4: Sections of regions with the origin inside and outside. The radial map assigns the direction of the displayed ray to a boundary point.

Example 14.7 (Degree of radial flux)**.** Let β_{S^2} be the outward area form on the target sphere and let $r = \|x\|$. For tangent vectors v and w at $x \in S$,

$$\mathrm{d}G_x(v) = \frac{v}{r} - \frac{x(x \cdot v)}{r^3}, \qquad (G^*\beta_{S^2})_x(v, w) = \frac{\det[x\ v\ w]}{r^3}.$$

Indeed, the target form is the determinant with first column x/r. Every additional term parallel to x in the other columns gives zero. This pullback is precisely Lecture 12's Coulomb flux form restricted to S. Since the target area is 4π, its flux calculation gives

$$\deg G = \frac{1}{4\pi}\int_S G^*\beta_{S^2} = \begin{cases} 1, & 0 \in \operatorname{int} K, \\ 0, & 0 \notin K. \end{cases}$$

The flux is therefore an integer multiple of the sphere's area.

For complex maps use $\mathbb{C}\mathrm{P}^1$ with its complex orientation: if $z = u + \mathrm{i}v$ is a coordinate, the ordered real coordinates (u, v) are positive. The transition $w = 1/z$ has positive real Jacobian away from zero. This orientation agrees with the outward Bloch sphere of Lecture 7: $z = \tan(\theta/2)e^{\mathrm{i}\phi}$ gives its ordinary angular parametrization. Equivalently, the Bloch map pulls the outward area form back to $4\mathrm{d}u \wedge \mathrm{d}v/(1 + u^2 + v^2)^2$, which is positive.

Example 14.8 (Powers on the complex projective line)**.** For $n \in \mathbb{Z} \setminus \{0\}$ consider the map $f_n(z) = z^n$ on $\mathbb{C}\mathrm{P}^1$. For $n > 0$ it fixes zero and infinity. For $n < 0$ it exchanges them. Using $w = 1/z$ near infinity, in the source or target as appropriate, the expression near zero or near infinity is a positive power $w^{|n|}$ or $z^{|n|}$. Thus the extended map is smooth at zero and at infinity.

Fix a finite nonzero target value. Its inverse image has $|n|$ distinct points, by the polar formula for complex roots. At each such point the complex derivative nz^{n-1} is nonzero. Multiplication by a complex number $a + \mathrm{i}b$ has real derivative matrix

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix}, \qquad \det = a^2 + b^2 > 0.$$

Each inverse-image point therefore has local degree +1. The signed-count formula gives $\deg f_n = |n|$. In particular, $z \mapsto 1/z$ has degree +1 on $\mathbb{C}\mathrm{P}^1$. On the equatorial circle the same formula has degree n instead: a negative exponent reverses its one-dimensional orientation. Conjugation $z \mapsto \bar{z}$ has real matrix $\operatorname{diag}(1, -1)$, so it reverses the complex orientation and has sphere degree -1.

Self-check. Identify the degree of a smooth map $f\colon S^1 \to S^1$ with Lecture 3's winding number. In particular, compute it for $f(e^{i\phi}) = e^{in\phi}$.

Answer. Use the normalized angular form $\alpha_{\mathrm{ang}}/(2\pi)$ on the target. A smooth angular lift ψ of the loop $\phi \mapsto f(e^{i\phi})$ has $\psi(2\pi) - \psi(0) = 2\pi\,\mathrm{wind}(f)$ by Lecture 3. Its pullback is $\psi'(\phi)\mathrm{d}\phi/(2\pi)$, so integration gives $\deg f = \mathrm{wind}(f)$. For the power map the lift is $\psi(\phi) = n\phi$, and the degree is n for every $n \in \mathbb{Z}$.

Summary. On a nonempty closed connected oriented manifold, top cohomology is measured by integration. Between such manifolds of equal positive dimension, degree is a homotopy invariant integer, computed by signed inverse images at a regular value. It gives the even-sphere hairy-ball theorem for smooth fields and the smooth Brouwer theorem. Poincaré–Hopf relates zero indices to the Euler characteristic. De Rham's theorem identifies cohomology with real period functionals on homology. Part II adds a metric, connections and physical fields to these topological constructions.

Reading. Lee, *Introduction to Smooth Manifolds*, second edition, Theorems 17.30–17.31, pp. 454–455; Theorem 17.35, pp. 457–458; Theorems 17.38–17.39, pp. 459–460; Theorem 18.14, p. 484. Milnor, *Topology from the Differentiable Viewpoint*, §6, pp. 32–35 (online scan).